

The thin-film photovoltaic market will produce the equivalent of 26 gigawatts (GW) by 2015 and will generate well over \$20 billion in revenues during that time frame.

Economics of scale enable significant decrease in cost. Amorphous silicon-based photovoltaic is slated to accelerate in the TFPV space for several years to come reportedly with \$1.3 billion in revenues this year that will grow to \$4.1 billion in the year 2014. Thin-film modules, especially Applied Materials' 5.7m² module, have an advantage over crystalline silicon (cSi) modules. Thin-film modules are ideal for utility scale applications.

What is the present cost per watt for solar power and how will it decrease in the future?

The current price for solar energy depends on how sunny the location is and the cost of capital. In general, it is below \$0.25/kWh. Investing in solar R&D will bring down its cost per watt. Applied Materials, with its leadership in integrated circuit chip equipment manufacturing, has delivered its best in class engineering expertise to the solar industry. Reducing the cost per watt for thin-film module manufacturing will make a significant impact on cost for clean electricity.

Applied Materials' products are aimed at reducing the cost per watt of energy produced, to become more competitive with current electricity generation. We intend to accomplish this goal by combining our large-scale manufacturing expertise in the semiconductor segment with our engineering and technical expertise.

Is National Solar Mission's (NSM) plan to generate 20 GW of power by 2020 a feasible plan? Is Applied Materials geared to meet its demand?

Setting a goal is imperative for every activity one undertakes, more so as it spells out an objective in detail. Though the target by NSM is challenging, it is not out of reach. In fact, Europe has a goal between 150 and 200 GW. To achieve this goal, however, there is a need for a solar ecosystem in India that will help achieve NSM's goal of producing 20 GW power by 2020. Several European nations, such as Germany, have developed a system for generating 2-3 GW of solar power every year. This model if replicated in India can yield phenomenal results. Even if India achieves about 50 per cent of the target, it will be great for the country.

What is the future for LED in India? Is Applied Materials interested in entering the LED market?

Lighting remains one of the least efficient forms of electricity used in the world today. As much as 20 per cent of the world's total electricity is used just to create light, yet most light bulbs are still highly inefficient. A typical incandescent lamp uses less than 10 per cent of the energy

it consumes to produce light. Compact fluorescent lamps perform better, but still waste 75 per cent of the energy consumed.

Applied Materials is developing equipment for solid-state lighting, light-emitting diodes (LEDs) which convert electricity into light without the need to activate a gas (fluorescents) or heat a filament.

Can you talk about Applied Materials' fab2farm model?

The Applied Materials fab2farm solar deployment model delivers low-cost solar energy and creates local jobs by linking a SunFab factory with a large utility-scale solar farm. Applied Materials fab2farm model creates jobs in communities, utilities and solar panel manufacturers to meet peak energy demand and achieve utilities renewable portfolio goals.

The aim is to accomplish a scale of the order of 100 MW to make cost-effective solar modules. A solar farm equipped with SunFab modules could be built for a price of \$3.50 per watt. Fabs result in creating jobs and businesses in a region, meeting the growing industry demands. By applying the principle of scaling, the costs come down and, at the same time, transport and logistic expenses are minimised because the farms are located closer to the market.

What kind of solar solutions would you recommend for India?

There is a huge demand for locally manufactured products which leads to an even bigger opportunity for domestic solar solutions. There are diverse and innovative solar products available in the market such as solar lanterns, solar home lights, solar-powered water pumps, grain grinders, crop dryers and Internet communication systems. These products are energy-efficient and cost-effective. Government and corporate India must spread awareness about these products as with a population of more than one billion India has a huge untapped market that can choose to use energy such as solar.

There are so many villages in India. Each household spends about \$ 5-8 a month on kerosene for lighting homes. Whereas, by installing a 70-watt solar panel, an entire household can generate electricity in less than \$ 5 instalment every month. This 70-watt solar panel on being charged for five hours can provide 350 watt hours of energy a day that can run an entire Indian household.

The most important solar solution for India is using solar energy to generate electricity. The cost for solar has reduced to a great extent and can resolve power shortage in the country—something that is witnessed especially during the summer season in India. Public-private partnerships will encourage investments in the solar sector in India which will enable the growth and consolidation of the solar ecosystem.